# The Entropic Difference-Body Inequality for Convex Measures Holds for Every Positive Excess Exponent

Run `fa_banach_001` — candidate full solution

August 11, 2026

**Abstract**

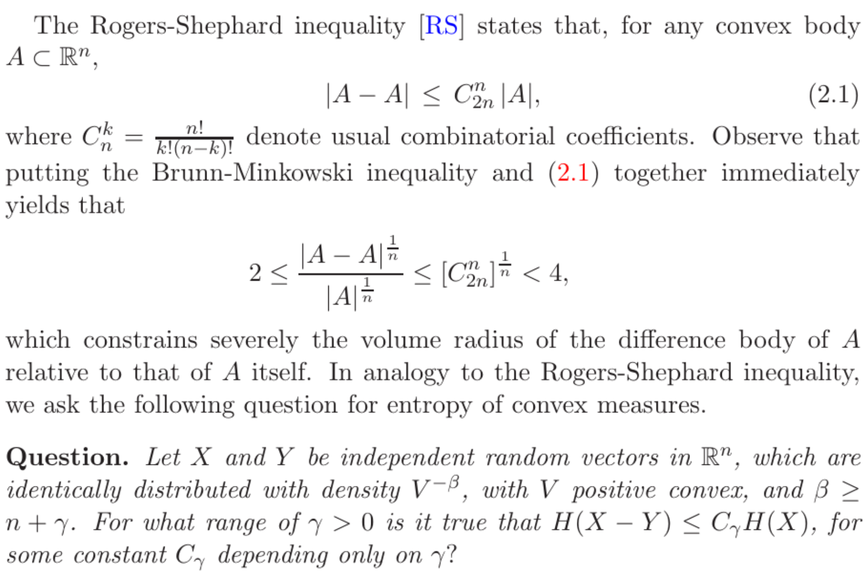
Bobkov and Madiman asked whether, for i.i.d. random vectors with density $f = V^{-\beta}$ on $\mathbb{R}^n$, the bound $H(X-Y) \le C_\gamma H(X)$ holds uniformly in dimension when $\beta \ge n+\gamma$. We answer affirmatively for every $\gamma > 0$. The proof quantizes the information content $-\log f(X)$ in bins of width $n$. Each bin lies in a convex superlevel set; Rogers–Shephard controls the support of every conditional difference. The sharp varentropy estimate for convex measures controls both the larger information bin and the discrete mixing cost. This yields a completely explicit, though unoptimized, constant depending only on $\gamma$.

## 1 Question and result

For an absolutely continuous random vector $U$ in $\mathbb{R}^n$, write

$$h(U) = -\int f_U \log f_U, \qquad H(U) = \exp\left(\frac{2h(U)}{n}\right).$$

The question in Bobkov–Madiman [1] is reproduced below.


The Rogers-Shephard inequality [RS] states that, for any convex body $A \subset \mathbb{R}^n$,

$$|A - A| \le C_{2n}^n \, |A|, \tag{2.1}$$

where $C_n^k = \frac{n!}{k!(n-k)!}$ denote usual combinatorial coefficients. Observe that putting the Brunn-Minkowski inequality and (2.1) together immediately yields that

$$2 \le \frac{|A-A|^{\frac{1}{n}}}{|A|^{\frac{1}{n}}} \le [C_{2n}^n]^{\frac{1}{n}} < 4,$$

which constrains severely the volume radius of the difference body of $A$ relative to that of $A$ itself. In analogy to the Rogers-Shephard inequality, we ask the following question for entropy of convex measures.

**Question.** *Let $X$ and $Y$ be independent random vectors in $\mathbb{R}^n$, which are identically distributed with density $V^{-\beta}$, with $V$ positive convex, and $\beta \ge n+\gamma$. For what range of $\gamma > 0$ is it true that $H(X-Y) \le C_\gamma H(X)$, for some constant $C_\gamma$ depending only on $\gamma$?*

Figure 1: The source question and its Rogers–Shephard motivation (arXiv:1111.6807, PDF page 5).

Define, for $\gamma > 0$,

$$A_\gamma = 2\max\left\{1, \sqrt{\gamma^{-2} + \gamma^{-1}}\right\}, \tag{1}$$

$$B_\gamma = 2\pi e\left((A_\gamma + \tfrac{1}{2})^2 + \tfrac{1}{12}\right), \tag{2}$$

$$c_\gamma = 1 + \log 4 + \frac{A_\gamma}{\sqrt{2}} + \log B_\gamma. \tag{3}$$

**Theorem 1.** *Let $X, Y$ be independent and identically distributed random vectors in $\mathbb{R}^n$ with probability density*

$$f = V^{-\beta},$$

*where $V : \mathbb{R}^n \to (0, \infty]$ is convex and $\beta \ge n + \gamma$ for some $\gamma > 0$. Then*

$$h(X - Y) \le h(X) + nc_\gamma. \tag{4}$$

*Consequently*

$$H(X - Y) \le C_\gamma H(X), \qquad C_\gamma = \exp(2c_\gamma). \tag{5}$$

*Thus the answer to the source question is:* every $\gamma > 0$.

No symmetry, centering, moment assumption, or convexity of the difference density is required.

## 2 Two inputs

Put

$$Z = -\log f(X), \qquad \sigma^2 = \operatorname{Var} Z.$$

Then $\mathbb{E}Z = h(X)$. The sharp varentropy estimate of Fradelizi–Li–Madiman [2, Corollary 4.4], specialized from $s = -1/\beta$, is

$$\sigma^2 \le \beta^2 \sum_{i=1}^{n} (\beta - i)^{-2}. \tag{6}$$

**Corollary 4.4.** *Under assumptions and notations of Proposition 4.2, we have*

$$\operatorname{Var}(\widetilde{h}(X)) \le \sum_{i=1}^{n} (1 + is)^{-2}. \tag{19}$$

*Proof.* By Lemma 2.1, we know that $\operatorname{Var}(\widetilde{h}(X_\alpha)) = L''(\alpha)$, where the tilted random vector $X_\alpha$ has density proportional to $f^{1-\alpha}$ and $L(\alpha) = \log \mathbb{E} f^{-\alpha}(X)$ is the logarithmic Laplace transform. By Proposition 4.2, we know that $L''(\alpha) \le c''(\alpha)$, where $c(\alpha)$ is defined in (18). Then the variance bound (19) follows by differentiating $c(\alpha)$ twice and setting $\alpha = 0$. □

Figure 2: The sharp $s$-concave varentropy input in its original notation (arXiv:1512.01490, PDF page 9). Substitution $s = -1/\beta$ gives (6).

For fixed $n$ and $1 \le i \le n$, the function $\beta/(\beta - i)$ decreases with $\beta$. Hence $\beta \ge n + \gamma$ and (6) imply

$$\frac{\sigma}{n} \le \frac{n+\gamma}{n} \left( \sum_{k=0}^{n-1} \frac{1}{(\gamma + k)^2} \right)^{1/2} \le A_\gamma. \tag{7}$$

For completeness, if $n \geq \gamma$, the prefactor is at most 2 and

$$\sum_{k=0}^{n-1} (\gamma + k)^{-2} \leq \gamma^{-2} + \int_0^\infty (\gamma + x)^{-2}\, dx = \gamma^{-2} + \gamma^{-1}.$$

If $n < \gamma$, the sum is at most $n/\gamma^2$ and the product in (7) is at most $2/\sqrt{n} \leq 2$. This proves the last inequality.

The geometric input is Rogers–Shephard: for every convex body $K \subset \mathbb{R}^n$,

$$|K - K| \leq \binom{2n}{n} |K| \leq 4^n |K|. \tag{8}$$

## 3 Information bins and conditional difference bodies

Let $Z'$ be an independent copy of $Z$ and define the integer-valued indices

$$J = \left\lfloor \frac{Z - h(X)}{n} \right\rfloor, \qquad J' = \left\lfloor \frac{Z' - h(X)}{n} \right\rfloor. \tag{9}$$

For $a \in \mathbb{R}$, set

$$K_a = \{x : f(x) > e^{-a}\}.$$

Because $f = V^{-\beta}$, $K_a$ is a convex sublevel set of $V$. Moreover,

$$|K_a| \leq e^a, \tag{10}$$

since $f > e^{-a}$ on $K_a$ and $\int f = 1$.

Condition on $J = j$ and $J' = k$, put $m = \max\{j, k\}$, and set

$$a = h(X) + n(m + 1).$$

The definition of the floor shows that both conditioned vectors lie in $K_a$. Their difference is therefore supported in $K_a - K_a$. A density supported on a set of volume $M$ has entropy at most $\log M$; thus (8) and (10) give

$$\begin{aligned} h(X - Y \mid J = j, J' = k) &\leq \log |K_a - K_a| \\ &\leq n \log 4 + \log |K_a| \\ &\leq h(X) + n\big(m + 1 + \log 4\big). \end{aligned} \tag{11}$$

Since $J \leq (Z - h(X))/n$, averaging (11) yields

$$\begin{aligned} h(X - Y \mid J, J') &\leq h(X) + n(1 + \log 4) + \mathbb{E} \max\{Z - h(X), Z' - h(X)\} \\ &= h(X) + n(1 + \log 4) + \frac{1}{2}\mathbb{E}|Z - Z'| \\ &\leq h(X) + n(1 + \log 4) + \frac{\sigma}{\sqrt{2}} \\ &\leq h(X) + n\left(1 + \log 4 + \frac{A_\gamma}{\sqrt{2}}\right). \end{aligned} \tag{12}$$

Here the equality uses $\max(u, v) = (u + v + |u - v|)/2$, and Cauchy–Schwarz gives $\mathbb{E}|Z - Z'| \leq \sqrt{2}\,\sigma$.

## 4 The discrete mixing cost

It remains to show that the information-bin index itself has bounded entropy. Let

$$U = \frac{Z - h(X)}{n}, \qquad R = U - \lfloor U \rfloor.$$

Since $0 \le R < 1$, $\operatorname{Var} R \le 1/4$. The triangle inequality in centered $L^2$ and (7) give

$$\sqrt{\operatorname{Var} J} = \sqrt{\operatorname{Var}(U - R)} \le \sqrt{\operatorname{Var} U} + \sqrt{\operatorname{Var} R} \le A_\gamma + \frac{1}{2}. \tag{13}$$

We record the standard integer-entropy estimate. If $L$ is integer-valued with variance $v$, and $Q$ is independent and uniform on $[-\frac{1}{2}, \frac{1}{2}]$, then the unit intervals supporting $L + Q$ are disjoint and

$$h(L + Q) = \operatorname{Ent}(L), \qquad \operatorname{Var}(L + Q) = v + \frac{1}{12}.$$

The Gaussian maximum-entropy inequality therefore gives

$$\operatorname{Ent}(L) \le \frac{1}{2} \log \left(2\pi e(v + \tfrac{1}{12})\right). \tag{14}$$

Applying (14) to (13) proves

$$\operatorname{Ent}(J) \le \frac{1}{2} \log B_\gamma. \tag{15}$$

The entropy-of-a-mixture inequality says

$$h(W) \le h(W \mid D) + \operatorname{Ent}(D)$$

for a continuous $W$ and a countable discrete $D$. Use it with $W = X - Y$ and $D = (J, J')$. Since $J, J'$ are independent, (12) and (15) imply

$$\begin{aligned} h(X - Y) &\le h(X - Y \mid J, J') + \operatorname{Ent}(J, J') \\ &\le h(X) + n\left(1 + \log 4 + \frac{A_\gamma}{\sqrt{2}}\right) + \log B_\gamma \\ &\le h(X) + nc_\gamma, \end{aligned}$$

where $n \ge 1$ was used in the last line. This proves (4); exponentiating $2/n$ proves (5).

## 5 Technical audit, scope, and novelty

A positive-probability bin is contained in a finite-volume convex superlevel set. Such a set is bounded up to closure, and (8) follows by the usual closure/approximation argument. Within each bin the conditional density is bounded and has bounded support, so the conditional entropies used above are finite. The countable mixture inequality follows directly from $g \ge p_d g_d$ for a mixture $g = \sum_d p_d g_d$, or by finite truncation. The hypothesis $\gamma > 0$ is precisely what is needed for $\beta > n$ in the varentropy theorem.

The source's maximum-density argument requires the difference density itself to be sufficiently convex. At $\beta = n + \gamma$, its entropy–maximum-density comparison loses order $n \log n$ on Pareto extremizers, so it does not answer the question. Later positional reverse-EPI results for $s$-concave densities do not state this fixed-additive-$\gamma$ conclusion. A bounded search of the run indexes, arXiv titles, exact formulas, keywords, and citations found [2] and other individual ingredients but no resolution of the displayed question. Novelty is therefore plausible but provisional: this packet should be reviewed as a candidate new full proof.

**Human-review recommendation.** Check especially the passage from the countable information partition to the mixture entropy bound. The direct inequality above makes this step robust; no limiting interchange involving differential entropy is needed.

## References


[1] S. G. Bobkov and M. M. Madiman, *On the Problem of Reversibility of the Entropy Power Inequality*, arXiv:1111.6807; in *Limit Theorems in Probability, Statistics and Number Theory*, Springer, 2013, 61–74.

[2] M. Fradelizi, J. Li, and M. Madiman, *Concentration of Information Content for Convex Measures*, arXiv:1512.01490; Electron. J. Probab. 25 (2020), paper 20, 1–22.

[3] C. A. Rogers and G. C. Shephard, *The Difference Body of a Convex Body*, Arch. Math. 8 (1957), 220–233.